DIC14
Schmitt Triggers

Specification-based laboratory report

17 October 2026

English edition
HSPICE S-2021.09 | Model library supplied separately

1 Objectives

Follow Lab 14 (PDF pages 58–61) to compare a CMOS inverter with the six-transistor Schmitt trigger, examine feedback sizing versus hysteresis/noise/delay, and test a distinct topology.

2 Specification mapping

Handout requirement (PDF page)	Implementation	Verification and scope
Step 1: Fig.14.1, three feedback sizes, bidirectional DC (58–59,61)	01_hysteresis, six cases	Independent sweeps avoid coupled branch jumps
Step 2: noise immunity (59–61)	02_noise_immunity	Same finite-duration disturbances
Step 3: speed comparison (60–61)	03_delay_comparison	Equal 8 fF and input edges
Question: alternate topology (61)	ALT; 05_alternative_dc	DC, noise and delay all tested
Additional numerical check	04_slow_ramp_check	1/10 μs ramps versus DC

3 Circuit and design derivation

The main series PMOS devices are each 3.6/0.25 μm and NMOS each 1.2/0.25 μm. Feedback PMOS/NMOS widths are 1.8/0.6, 3.6/1.2 and 5.4/1.8 μm for the three cases, all at L=0.25. The Fig.14.1 feedback connections are retained. The reference inverter uses widths 1.8/0.6 μm. The alternative is a strong input inverter ($W_p/W_n = 3.6/1.2$) driving two weak cross-coupled inverters (each 1.8/0.6). It also uses six MOS devices but a different feedback topology.

V^+ is the rising-input point where output falls through half supply; V^- is the falling-input point. Hysteresis width is $H = V^+ - V^-$. This state-dependent two-branch behavior must not be equated

with the two unity-gain margins of a single-valued inverter. The handout equations are square-law approximations; internal V_T and mobility are not independently verified here, so they are not treated as exact design equations.

4 Simulation setup

This teaching rewrite uses a separately supplied model at 1.0 V and 25 °C. The specified topology, loads and comparative dimensions are retained. The handout's 2.5 V process results are not acceptance values for this model. The chosen 1 V is a simulation condition, not a verified device rating. MOS bulk terminals are explicitly connected to the corresponding supply rails.

HSPICE S-2021.09 uses RUNLVL=6, retaining waveforms, printed data and native measurements. Figures are redrawn from actual data; CSV files are in each Lab's results directory. Successful solution does not establish process reliability. Warnings from encrypted model blocks remain suppressed; the existing diagnostic did not expose actionable further details, so the warnings are not declared harmless and complete model-limit compliance is not claimed.

DC sweeps 0–1 V and 1–0 V in 0.5 mV steps. Multiple .dc statements in the source did not reliably replace the intended reverse sweep; explicit start/stop/step parameters now define both directions. Parallel hysteretic instances also exhibited coupled Newton branch jumps, so DC uses one feedback case at a time. Bidirectional 1 and 10 μs/V transient ramps provide a cross-check. Noise stimulus matches Lab8; delay comparison uses 0.1 ns edges, 8 fF output loads and 50% crossings.

5 Results and specification checks

All five decks and 12 cases completed. Thresholds below are in V; ramp values use 10 μs/V.

Circuit	V_{DC}^{+}	V_{DC}^{-}	H_{DC}	V_{ramp}^{+}	V_{ramp}^{-}
SCH 1	0.5922	0.3968	0.1955	0.5928	0.3961
SCH 2	0.6332	0.3573	0.2760	0.6341	0.3564
SCH 3	0.6552	0.3353	0.3200	0.6565	0.3338
Alt	0.6321	0.0879	0.5441	0.6325	0.0873

Circuit	t_r (ps)	t_f (ps)	t_{pHL} (ps)	t_{pLH} (ps)
INV	106.95	51.92	63.86	88.77
SCH 1	219.10	84.35	149.36	203.59
SCH 2	260.52	101.05	196.67	230.75
SCH 3	301.50	118.33	244.97	256.13
Alt	427.75	106.61	70.70	311.56

6 Discussion

Stronger feedback increases DC hysteresis from about 0.195 to 0.320 V while increasing delay. The slower-ramp thresholds differ from independent DC by at most about 1.5 mV, a finite-speed and numerical check rather than proof of the zero-rate limit. A 0.6 V, 0.9 ns disturbance can switch the inverter while the Schmitt retains state during that interval. A short pulse near the DC threshold

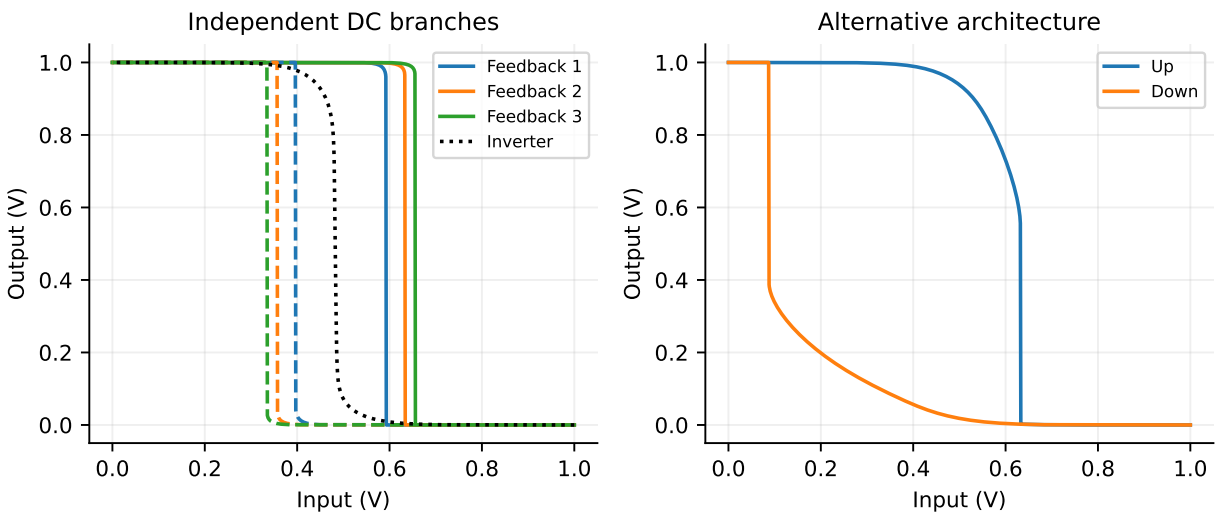


Figure 1: Independent DC hysteresis for three feedback strengths and the alternative; redrawn from this run's HSPICE data.

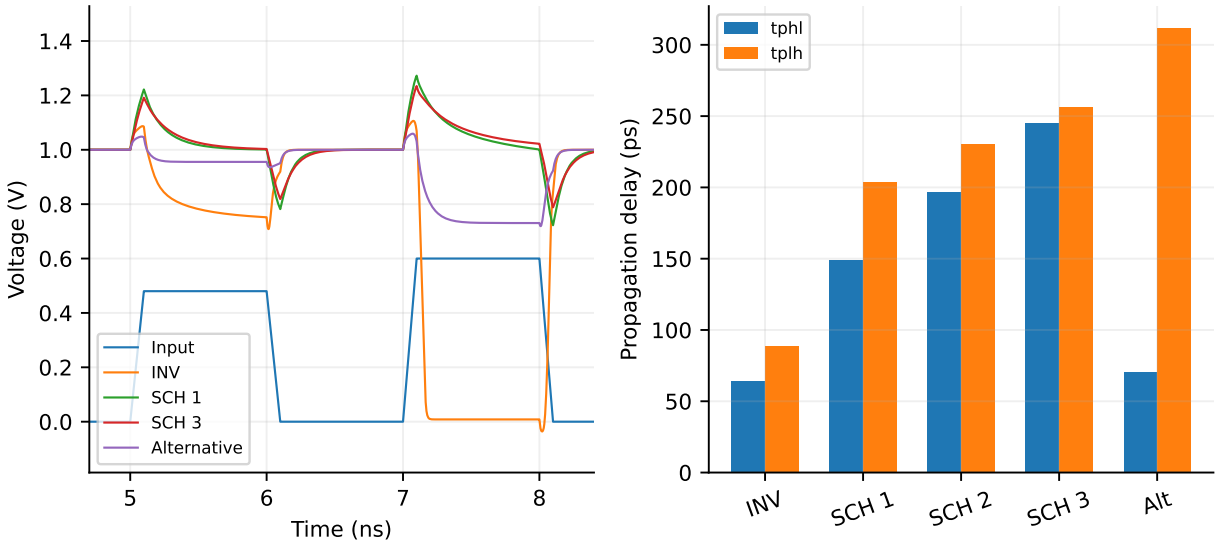


Figure 2: Finite-duration noise and propagation delays with equal loads; redrawn from this run's HSPICE data.

may not allow regeneration to finish; this does not establish a DC margin beyond the measured threshold. Actual data include temporary rail excursions whose reliability is unverified.

The alternative has about 0.544 V hysteresis but a falling-input threshold near 0.088 V, giving asymmetric noise behavior and a slower rising output. Larger hysteresis is therefore not universally better. Feedback strength must be evaluated together with speed, recovery conditions and input loading.

7 Conclusion

All three steps and the alternative-topology question have DC, transient and numerical cross-checks. The speed cost of stronger feedback and asymmetry of the alternative are explicit. Original-process voltage values and unverified reliability limits are not declared satisfied.